



Fatemeh Aghaei

Nationality: Dutch, Iranian

MSc Bioinformatics (Distinction) | DNA metabarcoding, high-throughput sequence analysis & reproducible pipelines | R, Python, Galaxy, Linux

Profile

Bioinformatics MSc graduate (Distinction) with hands-on experience in high-throughput DNA sequence analysis, reproducible R and Python workflows, and turning technical analyses into tools that non-specialists can run themselves. Four years in a GMP laboratory building automated data pipelines that connect people, data and systems. Comfortable on the Linux command line with Git version control, Galaxy and FAIR-aligned, reproducible analysis. Recently built a 16S rRNA metabarcoding pipeline in R with DADA2, and keen to bring high-throughput sequence analysis, reproducible pipelines and FAIR data to environmental DNA and biodiversity monitoring within the Data Competence Center at Naturalis.

Education and training

- 2023–2025 **MSc Bioinformatics**, *University of Birmingham*, Birmingham, United Kingdom, Grade: Distinction | Graduated Dec 2025.
- Hands-on high-throughput sequence analysis: NGS and RNA-seq data processing, quality control and downstream analysis using R and Python in reproducible, documented workflows.
 - Coursework in Applied Genomics & Next-Generation Sequencing, Statistical Machine Learning, metabolomics and multi-omics integration; built and ran analysis pipelines on Galaxy, with initial exposure to SnakeMake / Nextflow.
 - Thesis: investigated ARNT as a prognostic biomarker in non-muscle-invasive bladder cancer, integrating transcriptomic, CNV, clinical, survival and pathway analyses.
- 2017–2021 **BSc Cellular and Molecular Biology – Microbiology**, *Azad University, Tehran Medical Branch*, Tehran, Iran.
- Foundation in microbiology, molecular biology, genetics and evolution, including environmental and applied microbiology and biomedical laboratory science.

Working experience

- 2022–Present **Senior Microbiology Technician**, *Johnson & Johnson Innovative Medicine, QC-Microbiology*, Leiden, the Netherlands.
- Design and build automated data pipelines and reporting tools so non-technical lab staff can retrieve, visualise and report QC data themselves; initiated a Shiny-based QC reporting concept that scaled into a global Johnson & Johnson initiative.
 - Work with structured data platforms (Databricks), API / database connections and reproducible reporting (R, Quarto) to improve data accessibility, consistency and GMP compliance.
 - As a QC Digital & Technology Council representative, translate laboratory workflow needs into digital, automation-driven solutions across teams – bridging technical and non-technical colleagues.
 - Support GMP microbiology and CAR-T manufacturing through testing, data review, trend analysis and investigation support.

Scheepmakerspassage 20 – 3011VH Rotterdam, the Netherlands

📞 (+31) 64 410 48 29 • ✉ aghaei.fatemeh.1998@gmail.com • 🔗 [LinkedIn](#)

Portfolio

1/2

- 2026–Present **Volunteer Data Analyst**, *Statistics Without Borders*, Remote.
- Clean, curate and validate large structured healthcare datasets in R, producing reproducible, well-documented analysis and visualisation for a child-survival and hospitalisation-risk study.
- 2019 **Microbiology Laboratory Technician**, *Naft Hospital*, Tehran, Iran.
- Performed routine microbiological testing in a clinical laboratory, supporting accurate result documentation and reporting.

Selected projects

- 2026 **16S rRNA DNA metabarcoding pipeline (DADA2, R)**, *Self-directed project*, GitHub.
- Implemented a 16S rRNA amplicon metabarcoding workflow in R with DADA2 – quality filtering, error learning, denoising, ASV inference, chimera removal and taxonomic assignment – turning raw marker-gene reads into a taxonomically annotated ASV table.
- 2025 **Multi-omics analysis of pollutant stress in *Daphnia magna***, *MSc project – Computational Biology of Complex Systems*, R.
- Integrated transcriptomics (DESeq2) and LC-MS metabolomics (PCA, differential metabolites, KEGG / HMDB pathway enrichment) using RGCCA and DIABLO / mixOmics to characterise dose-dependent responses to environmental pollution in *Daphnia magna*, a freshwater bioindicator species; presented the work to the cohort.

Technical skills

- Programming & version control: R (advanced), Python, Git / GitHub, Bash; Quarto, LaTeX
- Sequence analysis & bioinformatics: DNA metabarcoding / amplicon (ASV) analysis (DADA2), NGS / RNA-seq analysis, quality control, differential expression, CNV analysis, pathway enrichment, survival & biomarker analysis, multi-omics integration
- Pipelines, Linux & reproducibility: Linux command line, Galaxy (hands-on); familiar with SnakeMake / Nextflow; cloud / remote compute (Google Cloud); reproducible, FAIR-aligned workflows
- Data, ML & integration: data wrangling, statistical modelling, machine learning, visualisation (Shiny, Tableau, ggplot2, PCA); structured data (Databricks), API / database connections

Selected courses & certifications

- 2023–2024 **Applied Genomics and Next-Generation Sequencing**, *University of Birmingham*, part of MSc Bioinformatics, Processing, integration and analysis of large-scale genomic and sequencing data.
- 2023–2024 **Statistical Machine Learning**, *University of Birmingham*, part of MSc Bioinformatics, Machine learning models for data-driven insight, predictive analytics and decision support.

Languages

English (Fluent), Dutch (Intermediate), Persian (Native). Dutch national, based in Rotterdam and already working on-site in Leiden.

Volunteering

- 2022–Present **Volunteer**, *Stichting NL Cares*, Amsterdam, the Netherlands.

References

Dr Roland Arnold, Group Leader, Dept. of Cancer & Genomic Sciences, University of Birmingham – r.arnold.2@bham.ac.uk

Dr Laura Grande, Process Microbiologist, Johnson & Johnson – laur.grande@gmail.com

Scheepmakerspassage 20 – 3011VH Rotterdam, the Netherlands

☎ (+31) 64 410 48 29 • ✉ aghaei.fatemeh.1998@gmail.com • 🔗 LinkedIn

Portfolio

2/2